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Cloud Leader · Builder · Podcaster



The On-Ramp Illusion:

Connecting to AWS, Azure, GCP

Megaport CONNECT
The Art of Modern Networking

OCTOBER 22, 2026 — CHICAGO

About Me



Let’s CONNECT.

https://wcollins.io/links

I have held the Trident

OPERATED — **Ran the circuits**

Nearly two decades of enterprise networks, cloud, and automation in healthcare and financial services.

BUILT — **Multi Cloud as a vendor**

Alkira, from lean startup to the Deloitte Fast 500 top 25 in North America. *(Acquired by Lumen)*

AUTOMATED — **Orchestration**

Director, evangelist, and IaC product development at Itential.

TODAY — **Architect and builder**

Director in Cloud, back inside the Enterprise.

Who's who on Olympus

APHRODITE
Developer

Ships beauty.
"It works on my laptop."

POSEIDON
Network Engineer

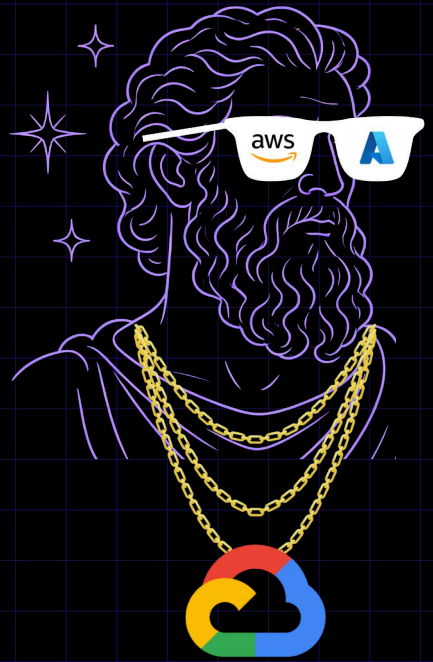
Salty. Not from the ocean. Holds the pager.

HERMES
Platform Engineer

Wants an API for everything. 40 tabs, all consoles.

ATHENA
Security

Wise. Strategic. Always wins.



Zeus
CEO, Olympus

"We are Multicloud now."

ACT I • THE ORACLE SPEAKS

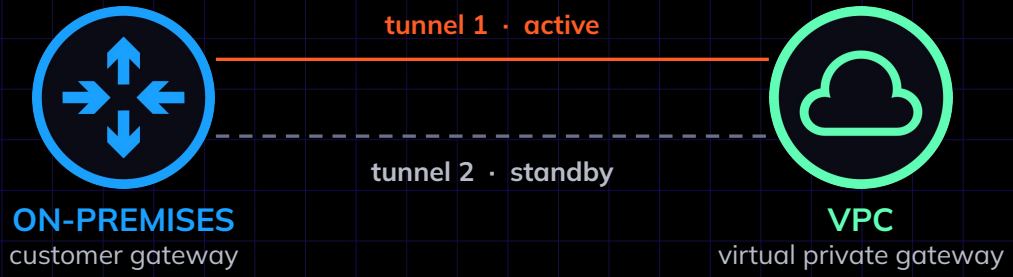
“Every green light is a prophecy.”

> The Oracle at Delphi

(Datasheet Division)



It started with one connection



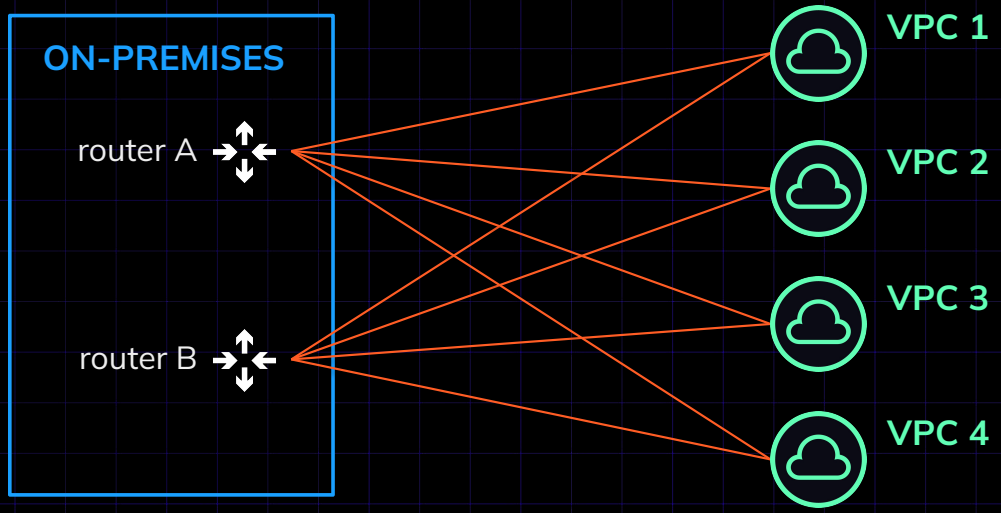
2 Tunnels per VPN connection

1.25 Gbps, up to per tunnel on a VGW

site-to-site VPN · IPsec over the internet

We've been building VPNs forever, we got this!

The Hydra > tunnels multiply



THE MATH

4 VPCs = 16 tunnels

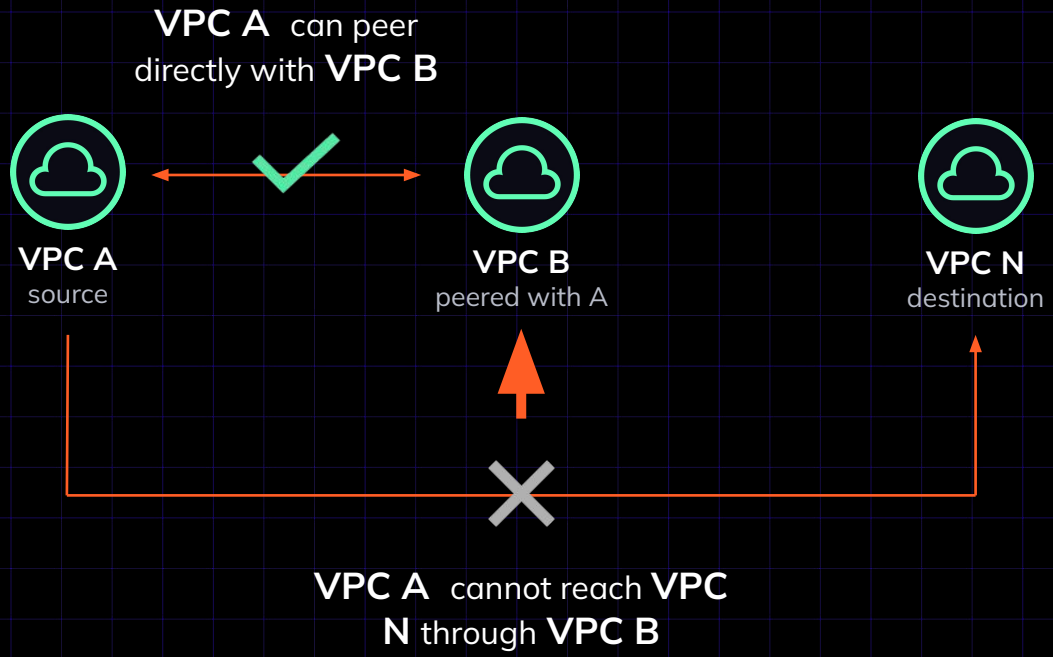
Two routers, two tunnels each. A BGP session on each one.

THE HAIRPIN

Spoke to spoke means spoke, internet, data center, internet, next spoke. Two tunnels for every flow.

Why so many VPCs?

Transitive routing > A cannot reach N through B



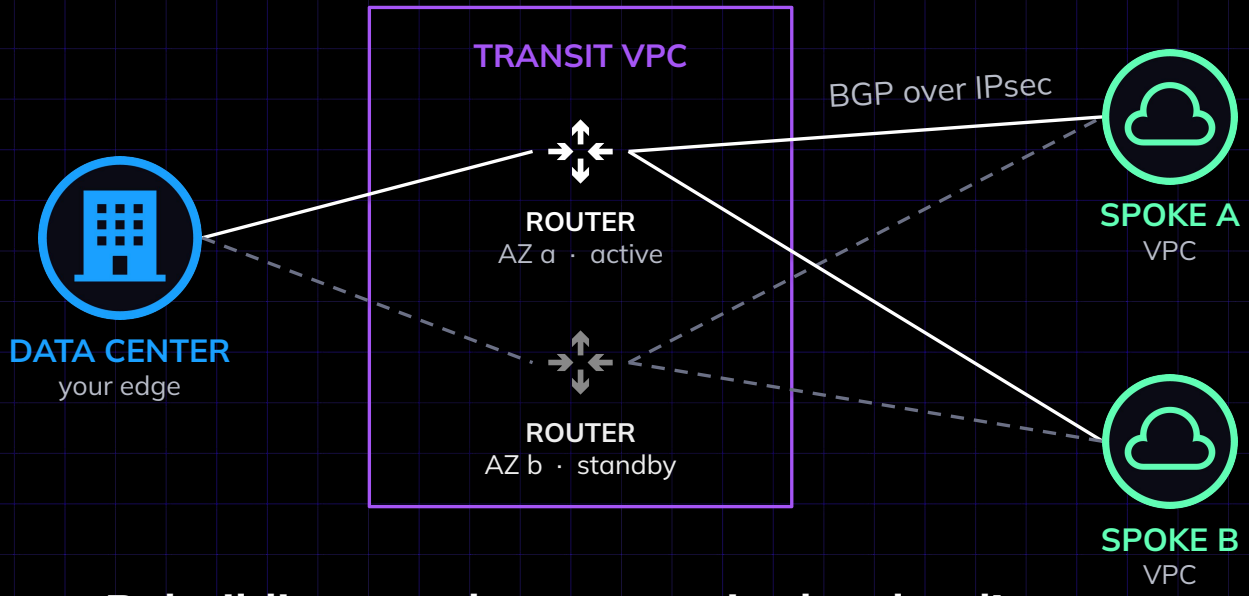
THE RULE

A peering connection is a point-to-point route between two VPCs. Nothing forwards across it for you.

WHAT IT ALSO MEANS

Six peerings for four VPCs, twelve routes by hand. Full mesh with no center, and no centralized control.

The transit VPC > lift and shift the network



THROUGHPUT
instance cap, halved by IPsec

BOTTLENECK
every spoke through one pair

PAID TWICE
egress billed at spoke and hub

LICENSES + PATCHING
you run routers again

FAILOVER
minutes on default timers

Rebuilding our data center in the cloud!
....One router at a time.

Direct Connect + TGW > The hub became a service



WHAT IT FIXED

Throughput, jitter, and tunnel sprawl. No router instances to license or patch.

WHAT IT DID NOT

Months of lead time, route limits that drop the session, and one cloud on the far end.

The hub was finally a service! For one Cloud...

Three clouds, three doors

ORDERED | OPERATED



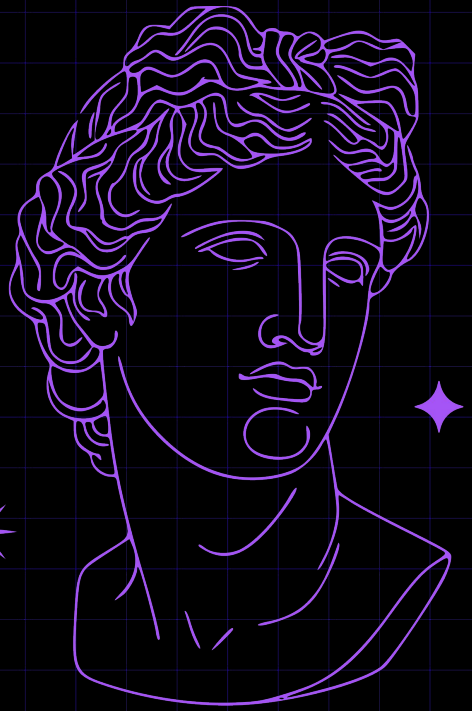
	CIRCUIT	ATTACHMENT	GATEWAY	NETWORK
AWS	Direct Connect	virtual interface	DX GW, then TGW	VPC and its route tables
AZURE	ExpressRoute	peering	ExpressRoute gateway	VNet and its UDRs
GCP	Interconnect	VLAN attachment	Cloud Router	VPC and its routing mode

ACT II • THE LABYRINTH

“You don’t choose Multicloud. It chooses you.”

> Homer, Ancient Greek Poet

(Would have probably worked in Enterprise Architecture)



One experience, three clouds

SAME EVERYWHERE

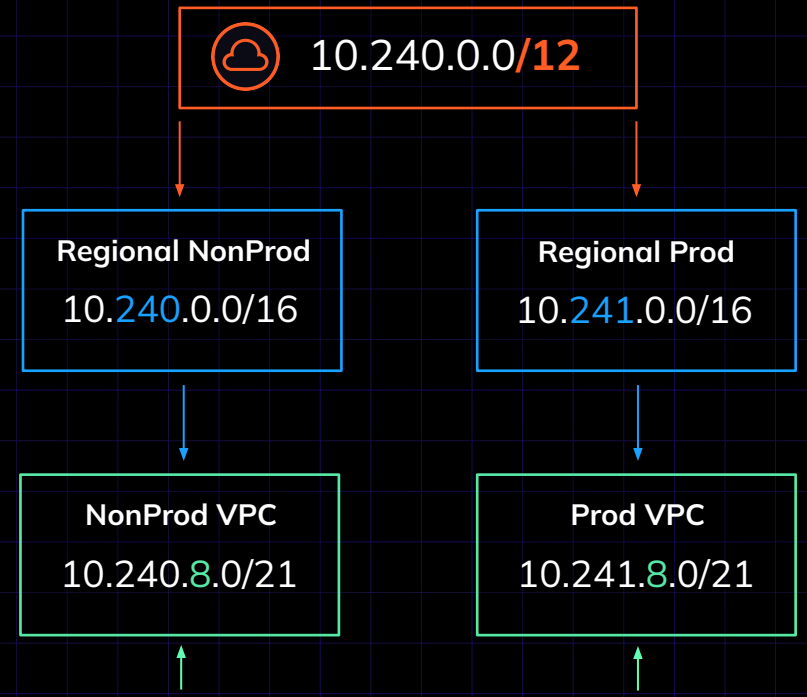
- One addressing scheme
- One security intent, three dialects
- One route policy, tagged on ingress
- One topology in every metro
- One failover drill, on a schedule
- One provisioning surface

PROVIDER-AWARE BY DESIGN

- Gateway objects and their scope
- Where security is enforced
- SLA math and route limits
- Each cloud's routing brain

Make it common where you can. Keep it provider-aware where you must.

Achilles' heel > addressing

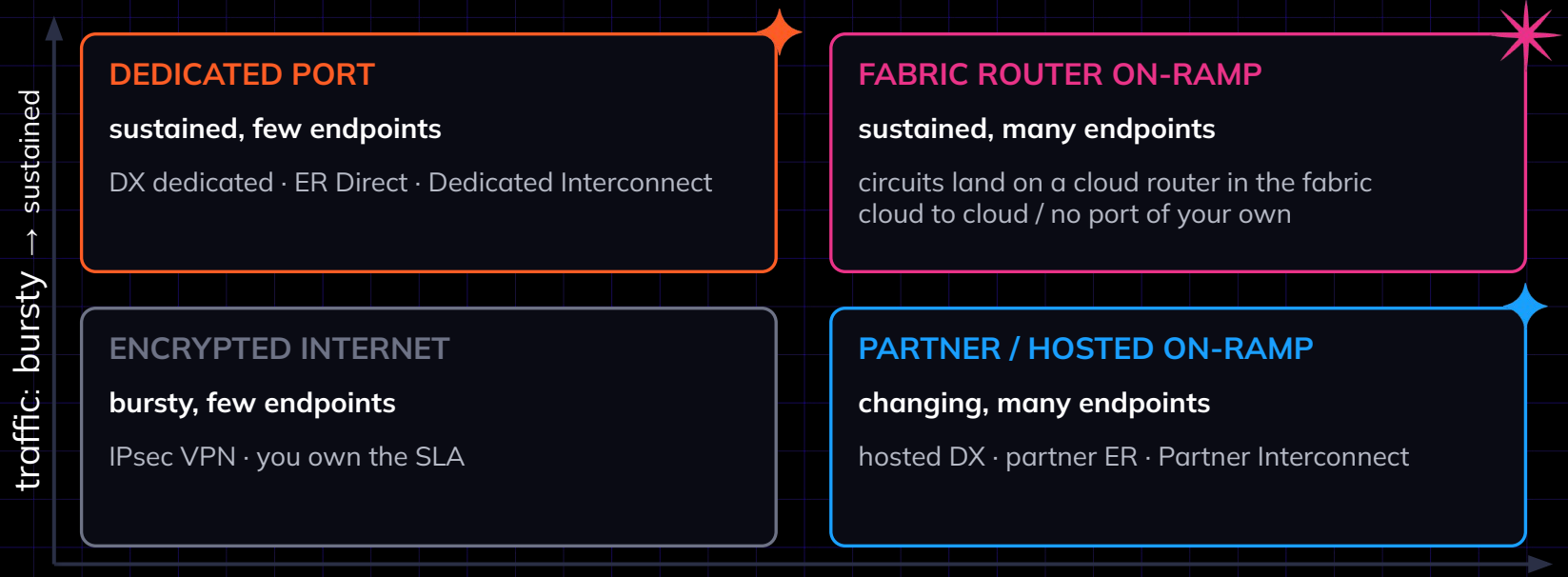


IP Addressing still matters in 2026...

Even numbered /16 allocated to NonProd and odd numbered allocated to production

3rd octet matches between NonProd and Prod VPCs.

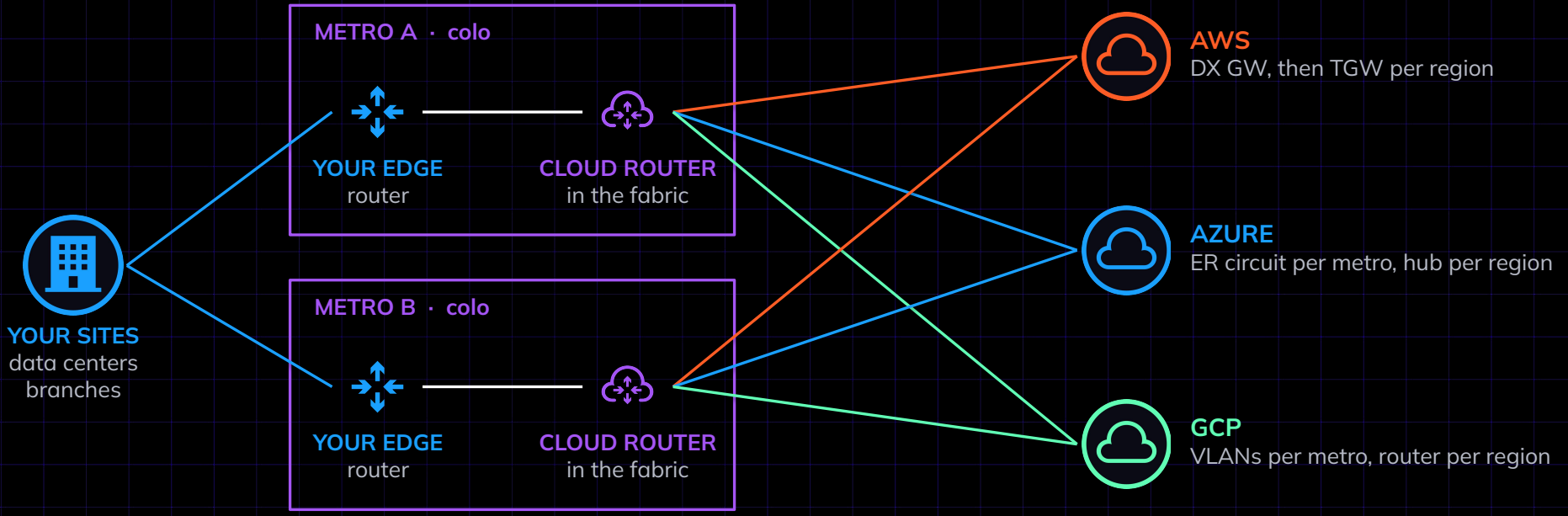
Choose the access model



(Whichever door you pick, the hub inside each cloud still does the fan-out: Transit Gateway · Virtual WAN · NCC)

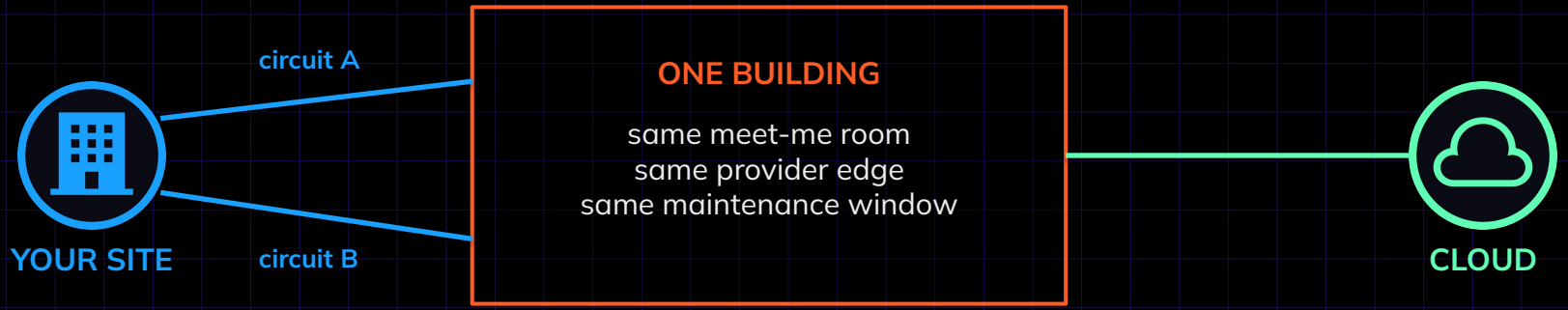
Start from the workload. Then pick the door.

The blueprint > two metros, a hub per region



Two metros. A hub per region. One policy surface.

Icarus > two circuits, one fate



AWS SLA
Four dedicated connections, two locations. Hosted: no SLA.

AZURE SLA
Two circuits, two peering locations. Metro is the middle tier.

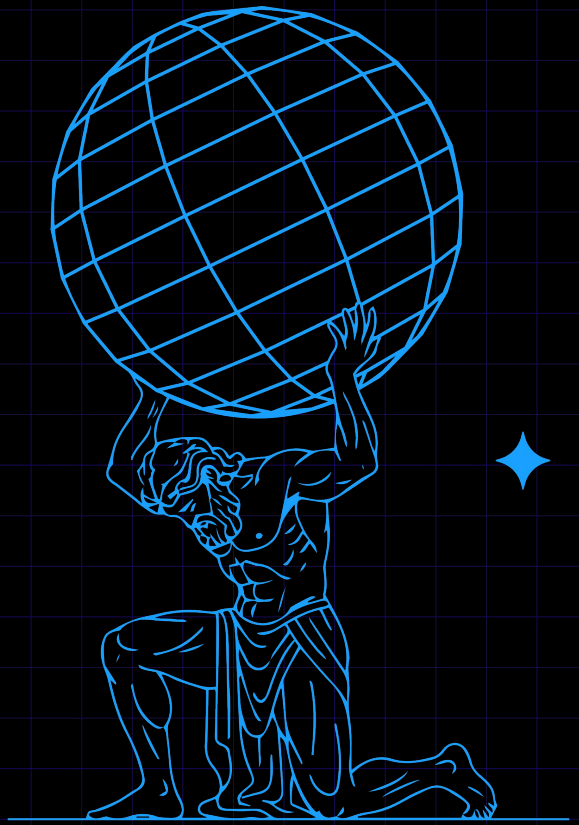
GCP 99.99%
Four attachments, two metros, two EADs, global routing.

Two lines. One fate. True diversity == failing separately

ACT III • ESCAPING WITH HERMES

“Declare it once. The pipeline carries it.”

> Hermes, Platform Engineering



Transit Gateway > the router is a resource



TRANSIT GATEWAY

regional in scope

A fully managed router that connects VPCs at enterprise scale.

ON-PREM	rack, cable, console, configure
CLOUD	declare, plan, apply

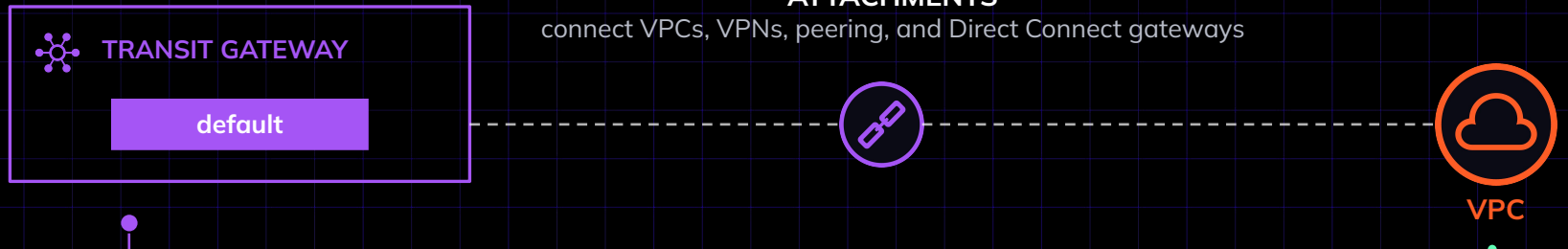
TERRAFORM

```
resource "aws_ec2_transit_gateway" "this" {
  description = "Managed by Terraform"

  tags = {
    Name = format("tgw-%s", var.region)
  }
```

The router became a resource block defined in code.

Attachments > the cable is code too!




```
resource "aws_ec2_transit_gateway_vpc_attachment" "this" {
  vpc_id             = aws_vpc.this.id
  subnet_ids         = aws_subnet.tgw[*].id
  transit_gateway_id = aws_ec2_transit_gateway.this.id

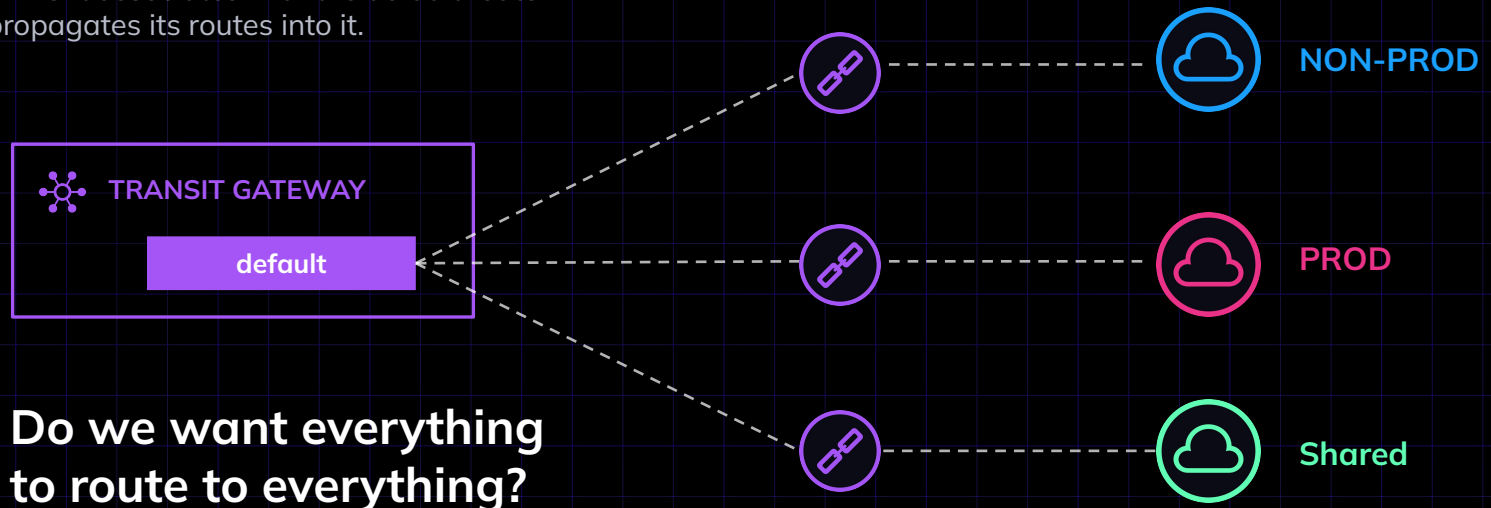
  tags = {
    Name = format("attachment-%s", var.vpc_name)
  }
```

An attachment is a cable you declare.

Default route table > any to any

OUT OF THE BOX

Every attachment associates with the default route table and propagates its routes into it.



Do we want everything to route to everything?

Custom route tables > segments as code

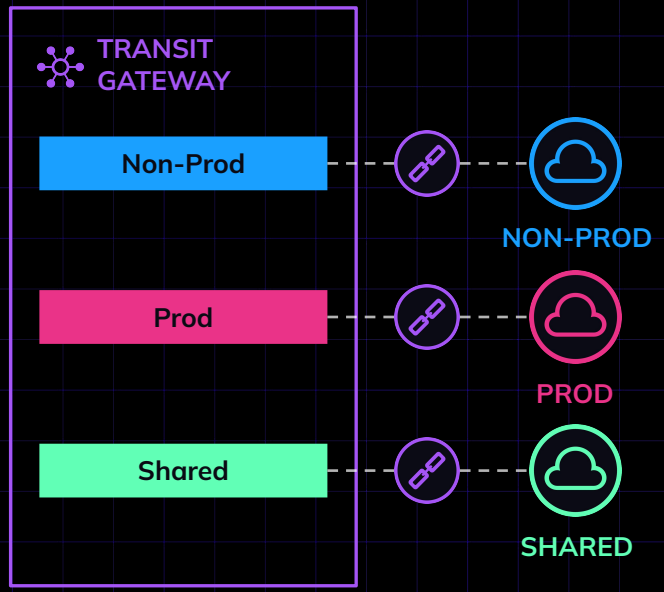
CREATE A ROUTE TABLE PER SEGMENT

```
resource "aws_ec2_transit_gateway_route_table" "prod" {
  transit_gateway_id = aws_ec2_transit_gateway.this.id
  tags = { Name = "tgw-rtb-prod" }
}
```

ASSOCIATE EACH ATTACHMENT WITH ONE

```
resource "aws_ec2_transit_gateway_route_table_association" "prod" {
  transit_gateway_attachment_id  = aws_ec2_transit_gateway_vpc_attachment.prod.id
  transit_gateway_route_table_id = aws_ec2_transit_gateway_route_table.prod.id
}
```

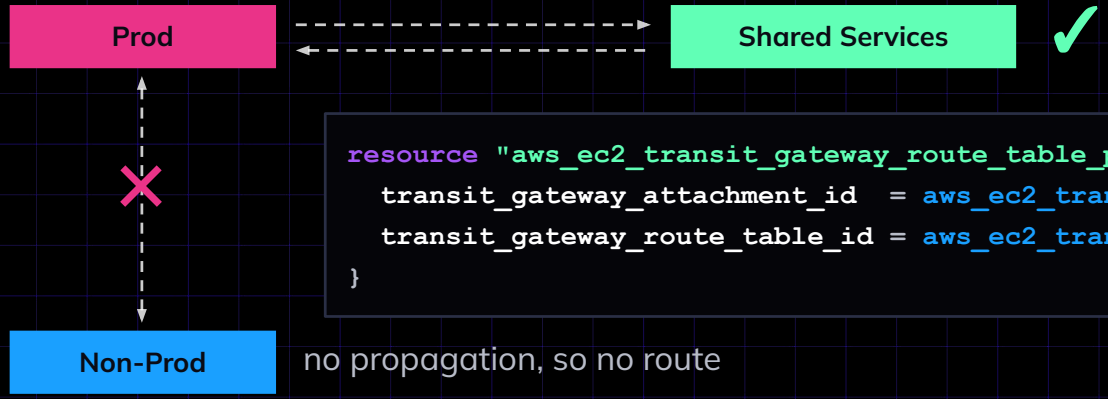
On the gateway, set default_route_table_association and default_route_table_propagation to "disable".



Transit Gateway route tables are the VRFs of the cloud.

Propagation > decide who talks to whom

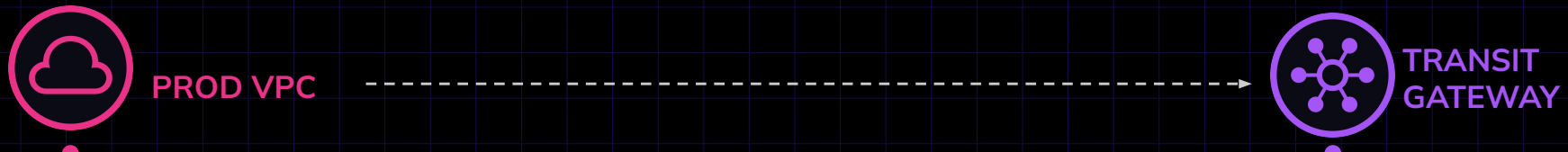
```
resource "aws_ec2_transit_gateway_route_table_propagation" "prod_to_shared" {
  transit_gateway_attachment_id  = aws_ec2_transit_gateway_vpc_attachment.shared.id
  transit_gateway_route_table_id = aws_ec2_transit_gateway_route_table.prod.id
}
```




```
resource "aws_ec2_transit_gateway_route_table_propagation" "shared_to_prod" {
  transit_gateway_attachment_id  = aws_ec2_transit_gateway_vpc_attachment.prod.id
  transit_gateway_route_table_id = aws_ec2_transit_gateway_route_table.shared.id
}
```

no propagation, so no route

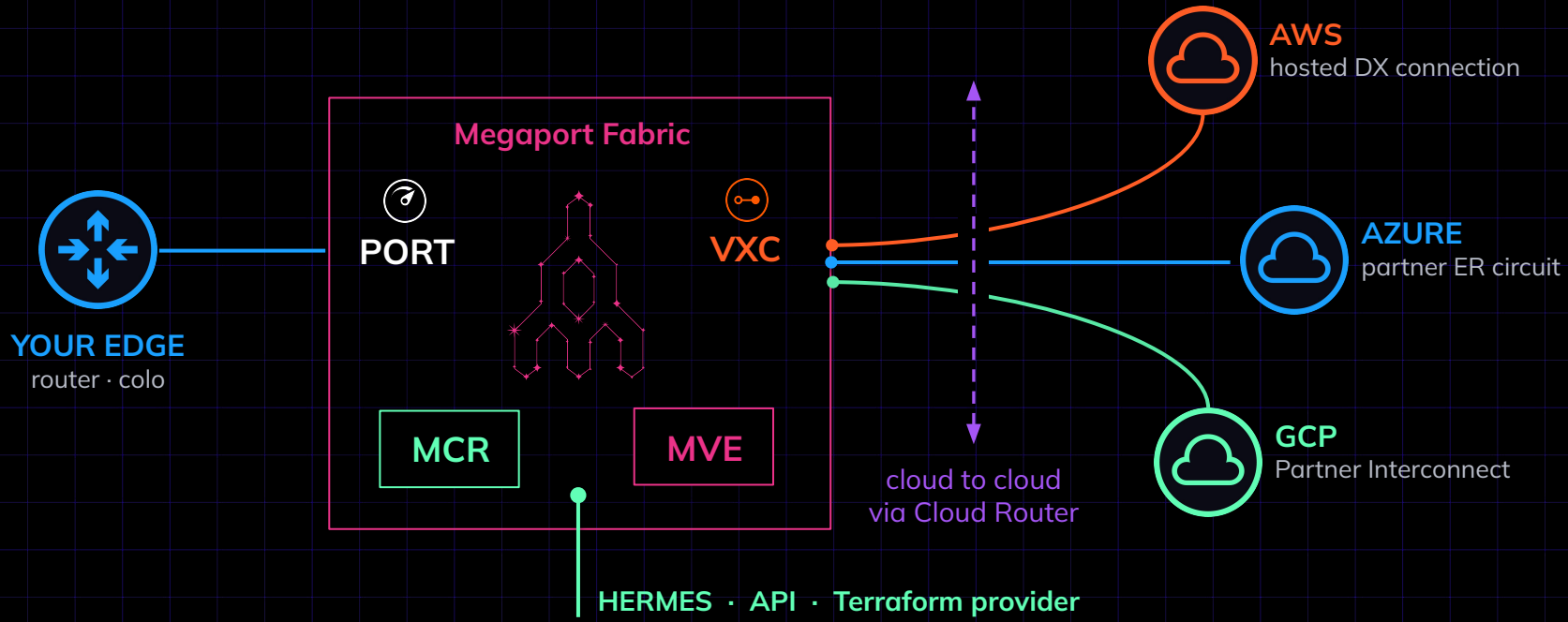
The VPC side > point the route at the hub




```
resource "aws_route_table" "prod" {
  vpc_id = aws_vpc.prod.id

  route {
    cidr_block         = "10.0.0.0/8"
    transit_gateway_id = aws_ec2_transit_gateway.this.id
  }
```

One Fabric, Three Doors



One provisioning surface. Three routing brains.

A sound operating model

Aphrodite shipped.

Poseidon slept.



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Thank You!

Q&A

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